

```

clc; clear
Fs=10000; Fp=1000; Fst=1500;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn);
freqz(b,a)
figure
% Magnitude & Phase
[h,n]=impz(b,a,50);
title('Impulse Response')
figure
stepz(b,a)
figure
zplane(b,a)
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);

```

